

COURSE NAME & CODE: ECA 0303 – Signals and systems

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Experiment 7.1

Reconstruction of Signal from Samples (Ideal Interpolation)

Aim: Reconstruct a bandlimited signal using sinc interpolation.

Program:

```
% Continuous time vector
```

```
t = -0.02:1e-4:0.02;
```

```
f0 = 300;
```

```
% Original continuous-time signal
```

```
x = sin(2*pi*f0*t);
```

```
% Sampling parameters
```

```
fs = 2000;
```

```
Ts = 1/fs;
```

```
% Sampled signal
```

```
n = -40:40;
```

```
ns = n*Ts;
```

```
xs = sin(2*pi*f0*ns);
```

```
% Signal reconstruction via sinc interpolation
```

```
x_rec = sinc_interp(xs, t, Ts);
```

```
% Plotting results
```

```
clf;
```

```
plot(t, x, 'k', 'LineWidth', 1.5);
```

```
hold on;
```

```
plot(t, x_rec, 'r--', 'LineWidth', 1.5);
```

```
hold off;
```

```
grid on;
```

```
legend('Original', 'Reconstructed');
```

```
xlabel('Time (s)');
```

```
ylabel('Amplitude');
```

```
title('Sinc Reconstruction');
```

```
% --- Sinc Interpolation Function ---
```

```
function y = sinc_interp(x, t, Ts)
```

```
    y = zeros(size(t));
```

```
    for k = 1:length(x)
```

```
        % Compute the unnormalized argument
```

```
        arg = (t - (k - 1 - length(x)/2) * Ts) / Ts;
```

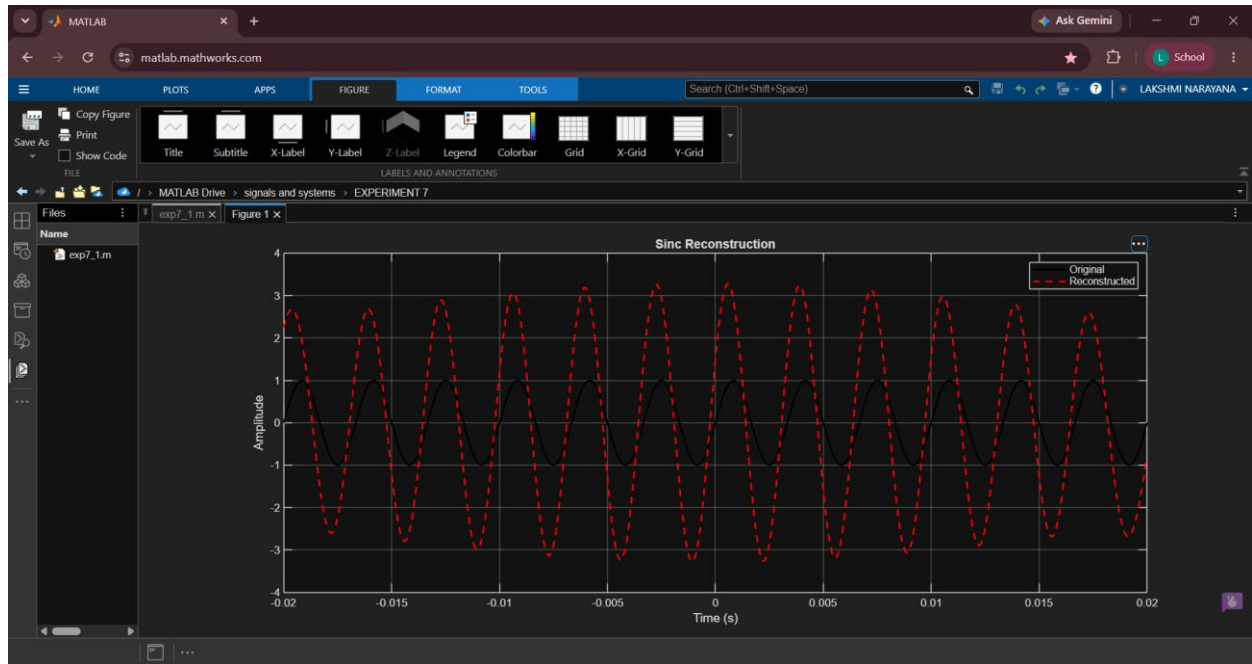
```
        % Divide arg by pi to adjust for MATLAB's normalized sinc(z) =  
        sin(pi*z)/(pi*z)
```

```
        y = y + x(k) * sinc(arg / pi);
```

end

end

Output:



Result:

Accurate recovery when sampling above Nyquist.